EXP 33: Prolog Program to Find the Number of Vowels

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Aim:

To develop a Prolog program that counts the number of vowels present in a given word using recursion.

Algorithm:

Step 1: Start the program.
Step 2: Read the given word.
Step 3: Convert the word into individual characters.
Step 4: Examine each character one by one.
Step 5: Check whether the character is a vowel.
Step 6: Increase the count whenever a vowel is found.
Step 7: Continue until all characters are processed.
Step 8: Display the total number of vowels.
Step 9: Stop the program.

Source Code:

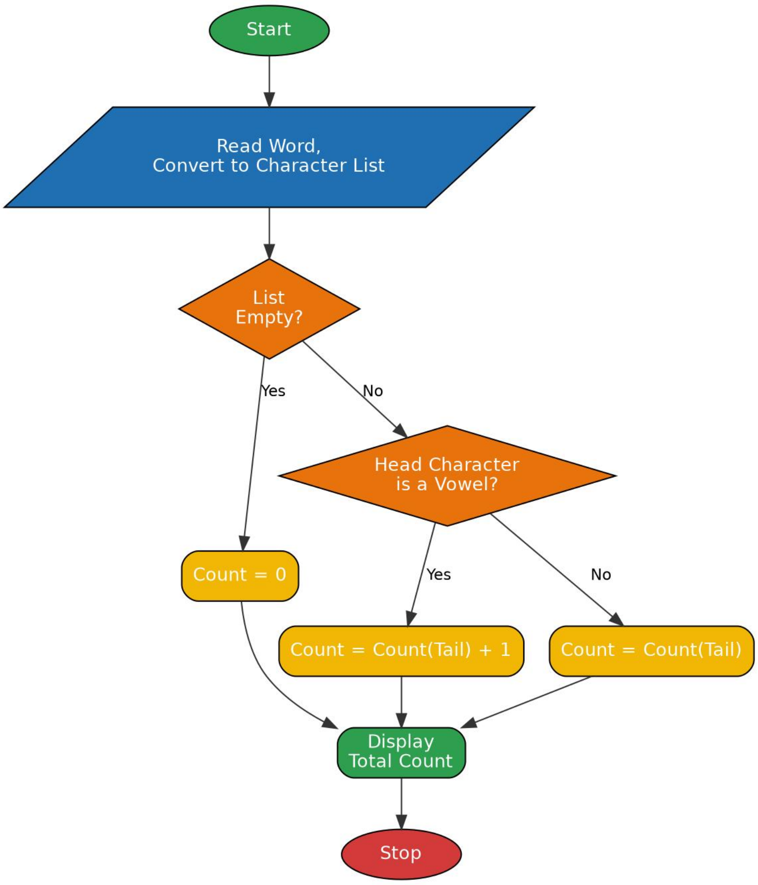
```
% Program to find the Number of Vowels

vowel(a).
vowel(e).
vowel(i).
vowel(o).
vowel(u).

count_vowels([], 0).
count_vowels([H|T], Count) :-
   ( vowel(H)
   -> count_vowels(T, Count1), Count is Count1 + 1
   ;  count_vowels(T, Count)
   ).

count_vowels_in_word(Word, Count) :-
   atom_chars(Word, Chars),
   count_vowels(Chars, Count).
```

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [vowels].
true.

?- count_vowels_in_word(education, Count).
Count = 5.

?- count_vowels_in_word(computer, Count).
Count = 3.

?- count_vowels_in_word(prolog, Count).
Count = 2.
```

Result:

Thus the Prolog program to find the number of vowels was written, executed successfully and the output was verified.